

Locale Framework Applied to Quantum Computing Innovations & Practical Applications

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Abstract

Why a reader should care: this paper turns every quantum-computing innovation and practical-application claim into a conditional truth with an explicit locale, then pairs each claim with the exact QPL 2026 paper that documents where it breaks. One table (the seams catalog T1-T8) replaces a year of vendor roadmaps: claims are conditional truths, the seams are public, and the energy exchange rate between paradigms is the currency. Premise-depth disclosure: the framework's unanalyzable primitives are 'locale' and 'interface' (imported from the companion record, Conditional Truths and the Locale Framework, DOI 10.5281/zenodo.21983324); the ontic structural realism reading is a named imported input (Ladyman-Ross), not derived; the energy exchange-rate theorems (arXiv:2510.08546, arXiv:2509.18854) are imported external results; the seams catalog is retrodictive with capped evidential weight; no dynamical theory of seams is claimed.

QNFO.UMP.011 companion artifact · 2026-08-17 · Applies 10.5281/zenodo.21983444 (v0.2) / latest 10.5281/zenodo.21983659 (v0.3) to the practitioner domain of quantum computing, with a QPL 2026 practitioner-paper review and challenge responses. All external claims verified against arXiv and the live QPL 2026 program (qpl2026.github.io/accepted/) on 2026-08-17. Research-cycle record: Phase 0 UIA (15/15, ZENODO-INQUIRY-1) → Phase 1 corpus sweep (KG 8,294 nodes / 1,634 papers; 3+ formulations × 8 hits; cross-system resolve; external arXiv verification) → Phase 2 (application, review, responses) → SO-WHAT-GATE.

0. Core claim (audited)

Co: The locale framework transfers to quantum computing. Every innovation and practical-application claim — “advantage”, “utility”, “fault tolerance by 2030”, “CV outperforms DV” — is a conditional truth with an implicit locale; the seams where claims fail to transfer are where the practitioner-relevant physics lives; the practitioner experiences a rendering (spec sheets, benchmarks, roadmaps), not the substrate; and the natural observables of the practitioner domain are counts and ratios, of which joules-per-solution is the composite scale primitive.

Premise-depth: the transfer imports UMP.011’s primitives (“locale”, “interface”); the structural-realist ontology is imported, not derived; the seams catalog is retrodictive (capped evidential weight); C₀ claims no dynamical theory of seams. The falsifiable core is UMP.011’s C3, instantiated where the seams are public record.

1. Conditional truths of quantum computing and their seams (the practitioner decision tool)

#	Claim (map)	Locale	Seam	QPL 2026 seam-docume (verified)
T1	CV quantum computing has a natural advantage	Unbounded energy, ideal measurements	Bounded energy + finite resolution → simulable by qudits; $\varepsilon \leq 1286 \cdot K \cdot n^2 \cdot E^{*2} / \sqrt{d}$	Maltesson–Rodung–Budinger–Ferrini–Calcluth arXiv:2510.08546
T2	More modes = more power	Mode count free	Modes trade against energy; constant modes → exponential energy	Brenner–Dias–Koenig, arXiv:2509.18854
T3	Advantage claims are device-level	Benchmark spec sheets	Exact classical Clifford+T sampling (2t evals); classical simulation of realistic odd-d GKP states	Koch (QPL EPTCS); Calcluth et al. (QPL)
T4	Statistics is a choice (paraparticles possible)	Higher-dim S _n reps	Quantum-permutation invariance forces Bose OR Fermi, model-independently	Mekonnen–Galley–Müller, arXiv:2502.17576
T5	Complex amplitudes are essential	Network experiments w/ source-state assumptions	Real QT reproduces ALL finite network correlations; RQT experimentally unfalsifiable	Hoffreumon–Woods, arXiv:2603.19208, 2504.02808; Ying et al. (QPL)
T6	Solovay-Kitaev (Archimedean)	R-metric approximation	Exact Clifford+R synthesis carries explicit	Deaconu–Gargava–Kalra–Mosca–Yard, arXiv:2510.11526

#	Claim (map)	Locale	Seam	QPL 2026 seam-docume (verified)
	governs synthesis		Bruhat-Tits building structure (arithmetic)	
T7	The transmon is a qubit	Two-level truncation	$d \approx 12$ levels; qubit = 1.9% correction term	QNFO The Two-Level Li (10.5281/zenodo.2148434)
T8	Fault tolerance by 2030	Roadmap rendering	10^2 – 10^3 physical-per- logical overhead $\times$ cooling budget	QNFO Physics of Computation (10.5281/zenodo.2125501) JPCUB P0 (10.5281/zenodo.2163702)

Map/territory (C2): T1–T3 are map failures of the same type as the photon-mass and spin catalog entries; the territory that transfers is energy accounting (joules per solution and components).

Rendering interface (C3): the practitioner’s first-person evidence is image-space; the substrate metric (Archimedean circuit model vs ultrametric/arithmetic structure) is not decidable from inside the interface — yet structure leaks: Buildings for Synthesis with Clifford+R (a Bruhat-Tits building inside a synthesis problem), ZW-calculus in q-arithmetic (Vilmart–Nguyen, QPL poster), and the re-proven arithmetic nature of Clifford+R. Each is a seam where the rendering fingerprints its construction (the C3 probe principle).

Scale primitives (C4): qubit/T-counts are counts; error rates and anharmonicity are ratios; advantage is a ratio of counts that is a function of the energy ratio. Joules-per-resolution is the composite primitive — externally theorem-backed by T1 and T2:
energy is the interconversion currency between paradigms.

2. Practitioner-paper review (QPL 2026, ~150 accepted, verified from the live program)

Tier 1 (practitioner): Buildings for Synthesis with Clifford+R (2510.11526) · Koch T-by-T sampling · Maltesson et al. CV↔DV (2510.08546) · Calcluth et al. GKP simulation · SpiderCat cat-state prep · Pauli Gadget Synthesis · Yamazaki–Akibue optimal T-count · Bian–Li–Ross–van de Wetering–Zhao multi-qudit Clifford rules (odd prime dims) · Vandaele comparators · ZX-Flow · Fault Tolerance by Construction · Backens–Perdrix flow rewrites · MWPM-preserving rewrites · DI-QKD single measurement · quantum-walker QRAM · Phantom codes · Transversal AND · Laakkonen distributed Clifford synthesis · Szyniszewski automated optimization · Sutcliffe treewidth/T-count hybrid · Kundu hardware-aware RL · magic-state cultivation (Wan–Zhong; Zurel–Davis stabilizerness).

Tier 2 (boundaries — challenge QNFO positions, see §3): Mekonnen–Galley–Müller · Hoffreumon–Woods pair · Ying foil theories · Lindenhovius–Zamdzhev operator spaces/linear logic · Arrighi dual-complex numbers · Baltag–Smets.

Adjacent practitioner venues: CWI Summer School on Quantum Algorithms & QEC, Amsterdam, Aug 24–28 (registered, order 101825); CWI QEC workshop Oct 28–30. (TQC 2026/QEC26 tracked via conference radar; no titles asserted without page verification.)

3. Challenge responses

CH-1 (Mekonnen–Galley–Müller, arXiv:2502.17576): parastatistics ruled out model-independently. Response: consistent with QNFO’s pre-registered F2 negative result — statistics is a locale effect of permutation invariance, not a theorem of any single formalism; the parsimony ledger (Boson/Fermion v1.5 §9, 10.5281/zenodo.21962904) gains an external upper bound; open probe: does complete invariance also bound the DHR-locality channel-count route, and does it constrain the 2D braid-group locale (anyons) at all? The anyon seam is untouched by the no-go.

CH-2 (Hoffreumon–Woods, arXiv:2603.19208/2504.02808; Ying foil theories): RQT empirically indistinguishable. Response: the exchange scalar $R = e^{(2\pi i s)} = (-1)^{(2s)}$ is real for bosons/fermions — QNFO’s statistics seam never required complex amplitudes; complex structure is load-bearing only at the anyon seam and in interference algebra. “Amplitudes are complex” is itself a conditional truth — true in the braiding locale, indistinguishable from a real rendering in every finite-network locale probed. The cleanest in-the-wild instantiation of C3.

CH-3 (Maltesson 2510.08546 + Brenner 2509.18854): energy exchange rate. Response: adopt — the first externally proven seam crossing, paid in energy; the first quantitative locale boundary ($\epsilon \leq 1286 \cdot K \cdot n^2 \cdot E^2 / \sqrt{d}$). *Converts JPCUB’s first-principles thesis into externally-theorem-backed instrumentation. Open: tightness (E^2 vs $1/\sqrt{d}$); soft vs hard binning in the SSD construction.*

CH-4 (Koch T-by-T; Calcluth GKP simulation): classical-baseline erosion. Response: each re-indexes the advantage locale (T3); the invariant is joules-per-solution — including the classical sampler’s own energy cost.

CH-5 (Buildings for Clifford+R, 2510.11526): confluence, not challenge — independent external confirmation that arithmetic/Bruhat-Tits structure is load-bearing at a practitioner seam. Adopt as independent verification of the valuation spine (UMP); labeled honestly: independent confluence, not QNFO-derived.

4. SO-WHAT-GATE

Why a reader should care: §1 is a locale-indexed map of every quantum-computing claim a buyer, investor, or engineer hears — each paired with the exact QPL 2026 paper documenting where it breaks. One table replaces a year of vendor roadmaps: claims are conditional truths; the seams are public; the energy exchange rate is the currency.

Premise-depth disclosure: primitives “locale” and “interface” imported unanalyzed (UMP.011); structural-realist ontology imported (Ladyman–Ross), not derived; seams catalog retrodictive; no dynamical theory of seams claimed.

5. Data-quality finding (cross-store DOI drift, flagged for the owning UMP.011 cycle)

Three stores currently carry three different DOIs for conditional-truths-locale-framework: D1 body_md frontmatter doi: 10.5281/zenodo.21983325 (v0.1) · search index DOI 10.5281/zenodo.21983444 (v0.2) · latest published record 10.5281/zenodo.21983659 (v0.3, P8R remediation 2026-08-17, commit 5248cfe). NEWVERSION-FRONTMATTER-CARRYOVER-1 class: the D1 body was not re-pointed to the latest deposited file's self-DOI. Remediation: verify the deposited v0.3 .md frontmatter == 21983659; re-ingest body_md from it; re-run Vectorize; verify all stores report one DOI.

Evidence (external verification, 2026-08-17, all live)

- arXiv:2502.17576v3 — Mekonnen, Galley, Müller, “Invariance under quantum permutations rules out parastatistics” (quant-ph/hep-th, 2025-02-24).
- arXiv:2510.08546v2 — Maltesson, Rodung, Budinger, Ferrini, Calcluth, “Equivalence of continuous- and discrete-variable gate-based quantum computers with finite energy” (2025-10-09).
- arXiv:2509.18854v1 — Brenner, Dias, Koenig, “Trading modes against energy” (2025-09-23).
- arXiv:2603.19208v1 — Hoffreumon, Woods, “Quantum theory based on real numbers cannot be experimentally falsified” (2026-03-19).
- arXiv:2504.02808v2 — Hoffreumon, Woods, “Quantum theory does not need complex numbers” (2025-04-03).
- arXiv:2510.11526v3 — Deaconu, Gargava, Kalra, Mosca, Yard, “Buildings for Synthesis with Clifford+R” (2025-10-13).
- QPL 2026 accepted program — qpl2026.github.io/accepted/ (fetched live; proceedings + talks + posters lists parsed).
- Corpus sweep — search_papers_enriched 3 formulations × 8 hits (Qubit Delusion series, Physics of Computation 21255013, JPCUB P0 21637028, QWAV strategy 21978952, UC-Langlands 20036379, p-adic anyon fusion 21208491, Two-Level Lie 21484345, Lifecycle of FTQC 18000790, No Thing There 21451776); KG stats 8,294/1,634; KG neighbors for paper:conditional-truths-locale-framework (BELONGS_TO prog-qinfo-ump; PRODUCED by proj-conditional-truths-locale-framework).